

MATLAB Based Graphical User Interface FOR DIGITAL IMAGE WATERMARKING

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A digital watermark is a kind of mark or image that is embedded in a multimedia carrier and can be used to verify the authenticity or integrity of the owner of multimedia carrier.

This graphical user interface (GUI) based watermarking system requires a login password for authenticity of the user. Without password

no one will be able to access the program and the GUI panel closes if a wrong password is entered. Therefore you need the correct password to operate this project.

For watermarking you require two images: one cover image that is to be watermarked and the other a logo or some other watermark image. The cover image to be watermarked may be selected from any of the images or pictures available in the directory using 'Select the cover image' button. Similarly, a logo to be embedded in cover image may be selected using 'Select the logo/watermark' button.

In order to run the application you need a 32- or 64-bit operating system with MATLAB software, a cover image and a logo/watermark image. Fig. 1 shows the flowchart of watermarking process used in this application. In this project, the GUI panel/program is quite user friendly. That's why some

buttons are disabled at the start of the program and enabled only when user finishes the first operation.

Software

The application program has been developed in MATLAB version R2013a. When the *main_file.m* file is run, a screen opens up, prompting the user to enter the password (refer Fig. 2). The correct password allows you to access the main GUI panel. But an incorrect password closes the GUI panel/program. The default password is *kcet* in this case.

The main parts of the GUI panel are described under Table I. Different PSNR and MSE values for the cover image (*college.jpg*) and logo (*logo.jpg*) at different bits are listed under Table II.

After selecting the cover image and logo/watermark, you have to give the bit position where the pixel value of logo has to be embedded in cover image. It is '1' for least significant bit (LSB) and '8' for most significant bit (MSB). The watermark image will be generated on pressing 'Assemble/Embedded watermark' button. The similarity between cover image and watermarked image is evaluated by calculating the peak signal-to-noise ratio (PSNR) and mean square error (MSE) values. The watermarked image generated with LSB gives better result, with maximum PSNR and minimum MSE.

Steps for testing

1. Launch MATLAB, open and run the file *main_file.m* program
2. A password window will appear on the screen. Type password *kcet*

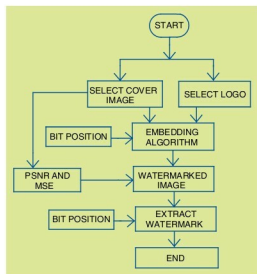


Fig. 1: Flowchart of watermarking process

TABLE I
GUI Panel Description

1. Select the cover image from your system.
2. Select the logo you want to embed on the cover image.
3. Select the bit position at which you want to embed the logo in cover image. 1 for least significant bit and 8 for most significant bit.
4. Click on to assemble your logo on the image.
5. Press button for PSNR and MSE to get values for quality evaluation.
6. Press Extract to get LOGO from watermarked image.
7. Information about the project.
8. Exit the working window.

TABLE II
Performance Evaluation of Cover Image (college.jpg) and Logo (logo.jpg)

Selected bit	PSNR value	MSE value
1	55.8658	0.16846
2	49.9471	0.658215
3	43.9186	2.63766
4	37.7413	10.9383
5	31.7062	43.8994
6	26.0804	160.34
7	20.3186	604.253
8	13.7929	2715.12